
Part III

Training-Methodological Considerations for Concurrent Aerobic and Strength Training



Methodological Considerations for Concurrent Training

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Introduction

As highlighted in earlier chapters, there are many studies that suggest performing both endurance and resistance training within the same training program (i.e., concurrent training) can lead to sub-optimal adaptations. However, although these studies provide evidence for interference to resistance training adaptations [i.e., reduced improvements in maximal strength, muscle hypertrophy, or power/rate of force development] with concurrent training [e.g., 1–6], these findings are not universal. For example, some concurrent training studies show no evidence for an interference effect [7, 8], while others suggest only certain adaptations are attenuated (e.g., compromised training-induced improvement in rate of force development, rather than attenuated maximal strength and/or muscle hypertrophy) [9]. On the contrary, there is also evidence for concurrent training further enhancing muscle hypertrophy (although, importantly, not maximal strength) [10–12] or post-exercise cellular signaling responses [13–15] compared with resistance training alone. The equivocal and varied nature of the response to concurrent training suggests the presence and magnitude of the observed interference effects might be related to methodological factors [6, 16]. These factors relate to training program design (i.e., exercise

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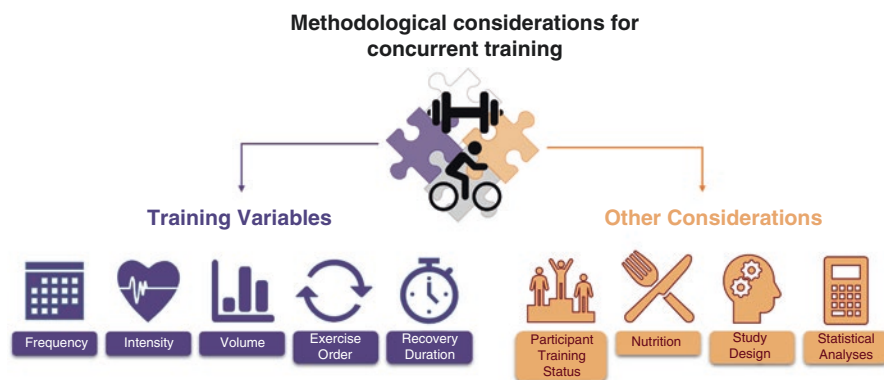


Fig. 13.1 Methodological considerations for concurrent training

frequency, intensity, volume, order, and recovery duration), as well as other factors including participant training status, nutrition, the study design, and statistical analyses used in the research (Fig. 13.1).

Training Program Design

Analysis of the results of studies investigating the *interference effect* suggests there are a number of training variables and other factors that need to be considered in the planning and prescription of concurrent training. Indeed, the volume, intensity, mode, and frequency of previous, current, and future training sessions, plus the exercise order and recovery duration between sessions, are all important factors that may influence the response to concurrent training. Furthermore, consideration of participant training status is critical when designing programs and interpreting concurrent training research, especially in relation to any observed interference effects. The following section outlines how these various methodological aspects should be controlled and/or considered when designing and interpreting concurrent training studies.

Training Status

Training status has been shown to affect the response to a single exercise session and plays an important role in determining subsequent adaptations to training. The concurrent training literature includes studies that have recruited sedentary [13, 17], recreationally trained [18–21], and exercise mode unaccustomed and/or accustomed participants [14, 21–24], with varying training outcomes often reported for different participant groups. For example, untrained participants elicit similar adaptive responses to both single-mode and concurrent training [24–26]. However, as

training is continued over a period of months to years, and the exercise stimulus is progressively increased to further enhance adaptation, the interference effect may become more apparent [27]. This suggests that in the untrained state concurrent training promotes predominantly generic, non-exercise mode-specific adaptations, while those with a greater training history exhibit more selective training responses to divergent stimuli, suggesting participant training status and history is a primary methodological factor to consider in the context of concurrent training.

In addition to sedentary, recreational, and trained participants, professional athletes may have a long training history, spanning several years, which is likely to impact on the planning of, and the adaptive responses to, concurrent training. Furthermore, the different competition schedules between individual and team sports will also impact planning and scheduling. For example, individual pursuit sports, which largely work on 4-year Olympic cycles, with specific yearly competition goals, may target specific phases in a year to develop specific physical qualities. When compared with team sports, in which competition may occur up to 4 times per week, the time available for training becomes a major issue, owing to the heightened need for recovery. From a logistical sense, changes to training schedules can also occur owing to injury management, weather, squad sizes, and available facilities. These “practical” issues, which are particularly prevalent across many sports, are likely to impact upon the acute and chronic adaptive responses to concurrent training.

Training Variables

Although there are many training variables that can be manipulated to influence the training response, the most common are frequency, intensity, and volume of training, as well as session order and the recovery between training sessions. While research evidence is limited, it is likely that these variables (individually and together) can influence the extent of any interference effect.

Frequency

Frequency relates to the number of training sessions (endurance and resistance) performed per week. While there appears to be little relationship between the frequency of resistance training sessions and improvements in running economy [28], $\text{VO}_{2\text{max}}$, or body fat [29], there are greater effects of endurance training frequency on resistance training adaptations. Indeed, performing ≥ 3 sessions of endurance training per week (which also results in a greater training status, and thus potential for greater interference) attenuates resistance training-induced adaptations [1–3, 30]. Conversely, performing ≤ 2 sessions per week appears to have a less negative impact on resistance training adaptations [8, 9, 31], which possibly also relates to the lesser trained status of the individual. As such, the frequency of endurance and resistance training sessions is an important methodological factor when programming concurrent training and when comparing the results of different studies.

Intensity

Training intensity is another important variable that likely influences concurrent training outcomes. This can be considered with respect to both endurance training intensity [e.g., high- vs. low-intensity continuous training, or high-intensity interval training (HIT) vs. moderate-intensity continuous training (MICT)], and resistance training intensity (e.g., high vs. low relative training intensities/loads). Endurance training intensity is a particularly relevant consideration, as higher endurance training intensities are a potent (and potentially time-efficient) stimulus for enhancing aerobic fitness and markers of cardiometabolic health [32]. For these reasons, HIT is often touted as a highly attractive strategy for improving cardiometabolic health and $\text{VO}_{2\text{max}}$, compared with traditional MICT [32, 33]. High-intensity exercise is also highly specific to the demands of many team and individual endurance-based sports, and so training modalities such as HIT often form an integral part of training programs aimed at enhancing performance across a wide range of athletes [34]. Given the popularity and effectiveness of high-intensity endurance training, the potential influence of this on the interference effect during concurrent training is an important consideration.

The intensity of endurance training performed within a concurrent training program may also influence the interference effect. Higher endurance training intensities are known to elicit greater residual fatigue of the trained musculature [35, 36] compared with lower-intensity endurance exercise [37, 38], and this can further attenuate force production capacity for at least 6 h after exercise [35, 36]. As discussed previously, this compromise in force production capacity can also negatively influence subsequent resistance training performance, an effect worsened following higher-intensity endurance exercise [38–40]. It is important to mention, however, that a number of additional variables also likely influence the neuromuscular demands of endurance training sessions, and therefore the likelihood for associated residual fatigue to negatively influence subsequent resistance training performance. Such factors include the endurance training modality (e.g., running vs. cycling), HIT work interval length, running surface (e.g., road, track, grass, or sand), running incline (e.g., hill vs. flat-ground running), and change-of-direction characteristics of running-based HIT [41].

A number of studies have demonstrated an interference effect with concurrent training incorporating a range of endurance training intensities, including low-intensity continuous training [4, 42], high-intensity interval training [43–46], or combinations of training intensities [1–3, 9]. There is, however, a lack of concurrent training studies directly comparing endurance training programs incorporating different training intensities [6, 47]. Such studies are required to determine the potential influence of training intensity on concurrent training outcomes. A study by Silva and colleagues [47] simultaneously investigated the effects of endurance training intensity (i.e., continuous vs. interval training) and modality (i.e., cycling vs. running) on neuromuscular adaptations to 11 weeks of concurrent training in physically active females. No differences for improvements in 1-RM leg press strength from baseline were found between training groups performing either resistance training only (~53%), or concurrent training incorporating either continuous cycling

(~39%), continuous running (~41%), or interval running (~47%). A limitation of that study, however, was that the endurance training protocols were only matched for total exercise duration, and not total work, making it difficult to infer the potential influence of training intensity on maximal strength outcomes [6].

Indeed, one methodological challenge when investigating the role of endurance training intensity in mediating the interference effect is that manipulating training intensity naturally influences total training volume, which in turn may confound study outcomes (*see paragraphs on training frequency and volume*). A potential solution to this problem is to equate the training volume of different endurance training protocols by matching them for total work performed, an approach used for comparing the influence of HIT and MICT on training adaptations [6, 48, 49]. Previous studies have observed attenuated maximal strength development with concurrent training programs incorporating either HIT [1, 43], MICT [4, 42], or combinations of both [2, 3]. Few studies have actually compared the influence of HIT and MICT as the endurance training modality on concurrent training outcomes [6], again making it difficult to determine the importance of endurance training intensity in mediating the interference effect. Fyfe and colleagues [6] compared the influence of work-matched HIT and MICT cycling on concurrent training outcomes, and observed almost identical interference of maximal (1-RM leg press) strength development, although HIT was more detrimental to improvements in various CMJ measures induced by resistance training (e.g., peak velocity and rate of force development) compared with MICT. The results suggested endurance training intensity per se may not be a critical factor for interference to maximal strength development, while higher endurance training intensities may be sub-optimal for the concurrent development of CMJ performance. Further work is required to better define the role endurance training intensity in modulating adaptation to concurrent training.

Volume

While training volume may be considered in the context of training frequency, it can also be depicted by the duration, equal also to work done, of a given training session and/or program. An increase in endurance session duration results in greater fatigue and substrate depletion [50], and therefore a requirement for longer recovery. Wilson et al. [29] observed a relationship between increasing daily endurance training duration and impaired resistance training-induced strength adaptations. This may indicate that increasing endurance training volume via the accumulation of total training time (and associated cumulative fatigue) would also impact on longer-term resistance training adaptations. Indeed, Hickson [2] only reported impaired strength adaptations in the concurrent training group from week 8 onwards of a 10-week training program. In that study, the concurrent group were performing endurance training (both running and cycling) 30–40 min per day 6 days per week—a volume of work that has not been matched in any other concurrent training study—on top of the 5 days per week of resistance training. This volume of training, which induces a large degree of cumulative fatigue, greatly superseded that of the endurance and strength-only training groups. More recently, Donges et al. [17] reported that when

concurrent exercise is matched for total work done with either resistance or endurance exercise performed alone (i.e., 50% of each mode combined), there are comparable increases in myofibrillar protein synthesis following concurrent and resistance exercise, and similar increases in mitochondrial protein synthesis between concurrent and endurance exercise. Taken together, this suggests that in sedentary, untrained participants, both aerobic and resistance exercise elicit a general additive effect on skeletal muscle exercise responses, while those with several weeks of training history are sensitive to both the volume of work done and the divergent exercise modes.

Contrasting that of endurance training, increasing resistance training volume in a concurrent training program through prolonged program duration (i.e., number of weeks) actually improves running economy in recreational ($\leq 55 \text{ mL kg}^{-1} \text{ min}^{-1} \dot{V} \text{ O}_{2\text{max}}$), well-trained ($55\text{--}65 \text{ mL kg}^{-1} \text{ min}^{-1} \dot{V} \text{ O}_{2\text{max}}$), and highly-trained athletes ($\geq 65 \text{ mL kg}^{-1} \text{ min}^{-1} \dot{V} \text{ O}_{2\text{max}}$) [28]. This is particularly important from the perspective of endurance sports, where running economy is a key performance indicator [51]. While the effect of resistance training session duration on concurrent training adaptations is less clear, there is evidence to suggest the ability to complete higher resistance training session volume, following endurance training, is compromised depending on the recovery duration between bouts [52] (see “*recovery duration proximity*” for detail). Another important “volume” factor to consider is that of accumulating resistance training volume. Given that higher resistance training volume results in greater muscle damage [53], and neuromuscular fatigue [54], it may be speculated that high resistance-training volume over time may impair the magnitude of concurrent training-induced adaptations. Collectively, training volume has varying roles in determining concurrent training adaptations and should be considered as a critical methodological factor when programming both endurance and strength training in a concurrent design, and when comparing the divergent findings in the literature.

Intra-session Exercise Order

Concurrent training offers a time-efficient alternative to single-mode training, particularly if both modes are performed within the same session. However, the acute interference hypothesis suggests residual fatigue and substrate depletion following a single exercise session may hinder the quality and performance of a subsequent bout, and induce unfavorable neuromuscular and molecular milieu, and thus compromise the potential for adaptation [16, 50]. Consequently, the choice of exercise order may be an important consideration for maximizing concurrent training adaptations. However, despite some evidence to support *acute* exercise order-dependent effects on strength [55, 56] and endurance exercise performance [57], as well as neuromuscular [58, 59] and molecular responses [19, 20], more research is needed to determine if, and how, these findings translate to order-dependent training effects. Most training studies to date report comparable gains in dynamic and isometric strength [60–70], power [65], hypertrophy [60, 62, 65, 67, 68], aerobic power and capacity [60, 62, 64, 67, 68], endurance performance [67, 69, 70], speed and agility [69, 70], irrespective of intra-session exercise order. This has been shown in a range of populations, including previously untrained/recreationally active men and women [60–63, 66–68], elite soccer players [69, 70], and elderly men [65]. Nevertheless, applying the existing research to

principles of training prescription, it would be prudent to recommend, where possible, the order in which concurrent endurance and resistance sessions are performed be dictated and periodized by the goals of that period in the training program.

Between-Session Recovery Duration

Increasing the recovery duration between divergent exercise modes might alleviate the potential negative effects of prior exercise on the quality of a subsequent session. Indeed, reductions in force generating capacity [35, 36], isokinetic strength [71, 72], and resistance exercise performance (i.e., number of reps completed) [52, 72] have been reported following endurance exercises of varying intensities and durations. These reductions in neuromuscular function and strength performance remained evident for at least 4–8 h post-endurance exercise, typically returning to baseline by 24 h [35, 36, 52]. Superior strength adaptations have also been observed following concurrent training sessions conducted on alternate days, compared to either same session [45, 73] or twice-daily training [73]. Collectively, these studies suggest that to restore muscle function for optimal strength performance and adaptation, at least 6–24 h recovery between concurrent sessions is warranted.

A greater recovery period between sessions may also benefit muscle growth. Indeed, Wilson et al. [29] reported a greater (albeit non-significant) effect size for hypertrophy when concurrent sessions were performed on alternate days (1.06) compared to the same day (0.80). However, others have shown similar gains in muscle size and mean fiber area regardless of whether sessions were performed on the same or different days [45]. Additional evidence in healthy, recreationally-active individuals suggests muscle hypertrophy and its associated acute molecular responses are not compromised, and are in fact potentiated, following short-term (5-week) training, irrespective of whether sessions are 15 min [12] or 6 h apart [11, 21]. A recent review by Murach and Bagley provides further support for comparable, if not greater, hypertrophic adaptations following concurrent training, compared with resistance-only training, regardless of the duration between modes [74]. Consequently, while muscle growth may not be limited by inter-session recovery duration, reductions in neuromuscular function and strength performance remain evident for 6–24 h. Increasing the rest period between divergent exercise sessions will allow for a greater quality of resistance training, increased force production and muscle fiber recruitment, and a better stimulus for strength adaptations.

Training Periodization

Athletic trainers may wish to manipulate training variables using a periodized approach, to match adaptations and performance with specific training and competition goals throughout a season. Since training variable manipulation clearly influences the interference effect [16, 29], the periodization strategy employed within a concurrent training program may also influence the degree of interference seen. Although studies directly examining the effect of training periodization on concurrent training adaptation are scarce [75, 76], some direct and indirect observations from the literature may provide some insight into the potential importance of concurrent training periodization. For example, given that higher endurance training

volumes or loads are associated with interference to strength training adaptations [29, 77], strategies to minimize endurance training volume during certain training phases may be beneficial. Periodized training approaches, whereby higher volumes/loads of endurance training are “cycled” between higher volumes/loads of strength training, may be useful for promoting improved strength training adaptations. Experimental evidence for the efficacy of this approach for minimizing interference is, however, currently lacking.

Research findings in world-class kayakers favored a block periodization approach, whereby specific endurance and strength components were trained simultaneously over multiple 5-week “blocks,” for improving kayaking performance [75, 76]. Compared with a traditional periodization model, the block periodization approach, which involved a 10% higher workload accumulation in each training block and less than half the total training volume, led to greater improvement in markers of kayaking performance. The authors noted that the choice of fitness component to be emphasized, and manipulation of training intensity in each block, were important considerations for any interference effect to be minimized [75]. Thus, for athletes in individual sports, block periodization may be an appropriate strategy, allowing endurance and strength parameters to be simultaneously developed towards a specific competition or goal. Team sports, however, involve a high frequency of competitive fixtures throughout long seasons, during which athletes are required to maintain high levels of strength, power, and endurance. Therefore, team sport athletes will likely benefit from a combination of different periodization strategies, dictated by the phase in the season, and the competition schedule. These strategies have been reviewed elsewhere [78]. In recreationally-active individuals, one study suggested that training periodization (specifically, endurance training intensity distribution) for this cohort may be less important, as both traditional and polarized distributions of training intensity and volume were equally effective for improving cardiovascular and neuromuscular fitness [79]. Nevertheless, this may reflect the potentially greater importance of endurance training volume, as opposed to training intensity, in mediating the interference effect [6, 29, 77]. Evidently, more research is needed to investigate the application of periodization strategies, with a specific focus on concurrent endurance and strength development, to identify strategies for minimizing interference during concurrent training.

Nutrition

Nutrition is another important methodological consideration, with strong evidence to suggest that nutrient availability can significantly modulate training adaptations [80]. Commencing exercise with low carbohydrate availability enhances metabolic and mitochondrial signaling responses [81, 82] and endurance adaptations [83–86], while protein ingestion (either as whole protein or amino acids) consumed pre- and post-exercise can augment muscle protein synthesis both in combination with and independent of exercise [87, 88]. Additionally, nutrient provision influences the expression of genes and proteins involved in protein breakdown [89], a physiological response regarded as unfavorable to achieving training goals. An equally

important, but less appreciated factor in terms of nutrition support is the effect of energy balance on resistance exercise-induced adaptations [90]. Indeed, Areta et al. [91] reported that following 5 days of energy deficit (30 kcal kg FFM⁻¹ day⁻¹), myofibrillar fractional synthetic rate is lower compared with being in energy balance. This suggests that an energy deficit induced by preceding endurance training (higher endurance training volume equates to greater energy deficit), which may exacerbate the endurance signaling response through carbohydrate restriction, may in fact decrease the anabolic response to the subsequent resistance exercise bout. Therefore, the independent effects of nutrient provision on divergent signaling and adaptive responses are important methodological considerations for those trying to maximize responses to both ends of the adaptation continuum.

To date, the majority of exercise-nutrition research has been conducted in single-mode exercise only [92]. Furthermore, within existing concurrent training literature, several studies have conducted training in the fasted state [13, 19, 20, 24, 93–95]. These practices are often unrepresentative of athlete practices, whereby carbohydrates and/or protein are consumed before, during, and after training to facilitate exercise capacity, recovery, and adaptation [96].

Study Design and Statistical Considerations

Even though there are roughly as many women in the world as men, women are notably under-represented as participants in the sport-science training literature. Women represent less than 40% of the participants in published studies [97], and this is likely to be even less in concurrent training studies. In this context, it is important to remember that women are not little men [98], and one shouldn't rely on research performed in men to inform concurrent training guidelines for women. For example, some research suggests there are differences in strength and hormonal adaptations between men and women following concurrent training versus strength training only [99]. Other data suggests women and men do not have the same recovery of strength in the 4 days following resistance exercise [100]. On the other hand, no sex-based differences have been observed for the muscle protein synthesis response to resistance exercise in the fed state [101]. Given the limited research to date, it is clear more research with women is required to better inform concurrent training guidelines for women. Furthermore, when women are recruited, they are generally studied only at their most biologically “male-like” (when neither ovulating nor menstruating). However, women need to be studied at all stages of their menstrual cycle as this is representative of the female training experience.

When women are recruited, it is often when small numbers of men and women are studied together. Given the established physiological differences between sexes [98], this has the unintended consequence of increasing the intra-participant variability and decreasing the power of the study to detect small differences between conditions. This low power is also characteristic of most of the concurrent training literature (even when only one sex is recruited), and means that small (and possibly important) effects of concurrent training may not always be reported. One solution is obviously to increase the sample size of concurrent training studies. However,

some researchers are also moving beyond traditional significance testing to provide confidence intervals, magnitude of effects, and the likelihood that an effect is harmful/trivial/beneficial [73]. These approaches are not without difficulties, as making an inference about magnitudes requires justification of the smallest worthwhile effect (which has not been established for all concurrent training variables). Another important statistical consideration is that while most concurrent training studies have traditionally reported mean responses, it is clear there is considerable individual variability in the response to training and researchers should consider reporting individual values for all variables.

Summary

Concurrent training is a common practice, and it is therefore important to understand any potential interference effects in response to this type of training. However, there are a number of methodological considerations when planning, and interpreting the consequences of, concurrent training. These include training program design, nutritional practices, the participant group studied, and the statistical analyses used in the research. While more research is required, it appears the interference effect is more likely to be observed in those with a longer training history, when performing ≥ 3 sessions of endurance training per week, and when there is a high training volume. Intra-session order is also important, and scheduling priority should be assigned to whichever mode reflects the primary training goal so as to maximize the quality of the exercise session and consequently the stimulus for adaptation. Increasing the recovery duration between sessions should also allow for a greater quality of subsequent training and thus a better stimulus for adaptation. Further work is required to better define the role of endurance training intensity on modulating adaptation to concurrent training. More research is also required to better inform concurrent training guidelines for women. As some of the potential interference effects are likely to be small, there is also a need for researchers to consider how they can increase the power of concurrent training studies. Finally, there is a need to better understand the mechanisms mediating the interference effect in order to inform strategies for maximizing concurrent training outcomes.

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